

Deepan Kulandaisami

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Education

National Institute of Technology Karnataka (NITK)

B.Tech in Electrical & Electronics Engineering; Minor in Mathematical & Computational Sciences

Mangalore, India

2020 – 2024

Technical Skills

Programming & Backend: Python, FastAPI, Django, Django REST Framework, REST APIs, WebSockets, SSE, SQL, PostgreSQL

AI/ML Engineering: LLM Applications, AI Agents, RAG, LangGraph, LangChain, CrewAI, Embeddings, Vector Search, Model/API Integration, RAG Evaluation, Prompt Evaluation

AI/Data Pipelines: Document Processing, PDF Extraction, OCR, Chunking, Embedding Pipelines, FAISS, Milvus, Qdrant, Pandas

MLOps/Cloud: Docker, Azure Container Apps, Azure Database for PostgreSQL, AWS S3, Git, Linux, Application Deployment, API Testing, Logging, Debugging

Distributed Systems: Celery, Redis, Background Jobs, Long-Running Workflows, Streaming Responses, WebSockets, SSE

Engineering: System Design, Testing, Root-Cause Analysis, Failure Analysis, Code Reviews, Agile/Scrum

Security: Keycloak, Azure AD/MSAL, Authentication, Authorization

Professional Experience

Associate Software Developer – AI Full Stack

Apr. 2025 – Sep. 2026

BONbLOC Technologies Limited

- Delivered **multiple production AI applications** across HR, legal, sales, research and compliance using **Python, Django, FastAPI, LangChain and LangGraph**, reducing manual business processes by **up to 60%** from architecture and development through deployment and production support.
- Built the company's first **AI-powered HR Assistant** for a **232-employee organization**, combining **RAG, Odoo HR APIs and tool calling** for employee policy queries and HR self-service; achieved **>99% faithfulness**, **>95% retrieval precision** and **>95% answer relevance** on RAG evaluation.
- Engineered a reusable **document-to-RAG platform** using **Docling, OpenAI embeddings and Milvus**, supporting complex policy documents, versioned knowledge bases and role-aware retrieval; achieved a **96% reduction in hallucinations** over vanilla LLM baselines while cutting query latency by **40%**.
- Developed **agentic proposal, RFP and document automation** using **FastAPI and LangGraph**, featuring stateful workflows, web search, SSE streaming and automated **PDF/PPT/DOCX** generation; achieved a **94% task success rate** on benchmark testing and reduced manual deliverable preparation effort by **75%**.
- Built production AI assistants integrating **multi-agent workflows, web search and Spring Boot REST APIs**, achieving **>95% tool-use precision** via OAuth-secured tool execution and MCP-style interfaces; extended core products with conversational and **voice-based interactions** using **OpenAI Whisper, Sarvam AI TTS and Sophrano TTS** maintaining a latency of **<450ms TTFT**.
- Developed ML-driven analytics for the company's **CSK IPL partnership** using **Logistic Regression, Random Forest and Gradient Boosting** across player-, match- and ball-level data; achieved **>88% prediction accuracy** on match-outcome modeling to support player evaluation and opposition analysis.

Assistant Engineer

Aug. 2024 – Mar. 2025

Petronet MHB Ltd. (PMHBL)

- Pipeline operators needed clearer visibility into pumping operations and system status. Developed web applications using **JavaScript, Node.js and Express.js** to visualize petroleum pipeline operations, pumping data, ullage information and Leak Detection System status. Improved access to operational information through web-based monitoring dashboards.
- Operational datasets required processing and visual analysis for monitoring and investigation. Built data-processing and visualization workflows using **Python, Pandas and Matplotlib**. Converted raw operational data into analytical visualizations that could be used during monitoring and incident analysis.
- Issues in mission-critical pipeline systems required timely investigation using multiple operational data sources. Worked with **SCADA, Leak Detection System (LDS) and Pipeline Intrusion Detection System (PIDS)** data for monitoring, troubleshooting and incident analysis. Supported operational teams during troubleshooting and investigation of pipeline-system issues.
- Collaborated with Cisco networking vendors on infrastructure troubleshooting and developed practical experience with TCP/IP and OSI-model concepts. Supported diagnosis of networking and infrastructure issues affecting operational systems.

Projects

Autonomous AI Portfolio – Digital Twin & Contact Concierge | *FastAPI, Python, BM25 RAG, Groq LLMs, Multi-Turn Agents, SMTP*

- Built an autonomous **AI Contact Concierge agent** featuring multi-turn conversational intake, structured JSON schema entity extraction (recruiter profile, company, intent), and automated tool execution via asynchronous SMTP email dispatch to inbox.
- Engineered an interactive **AI Digital Twin** using FastAPI and BM25 contextual RAG to answer technical architecture inquiries, strictly grounded on verified project metadata with hallucination guardrails and sub-second Groq LLM inference.

- **GitHub:** github.com/Deeps72-ux/Portfolio-website

QueryGenie – SQL Agent | *FastAPI, LangGraph, SQLGlot, LLMs*

- Built a natural-language-to-SQL chatbot that lets users query a database using plain English, with SQL validation and correction.
- **Live:** querygenie-nl-sql-studio.onrender.com
- **GitHub:** github.com/Deeps72-ux/querygenie-nl-sql-studio

NexusGraph – Multi-Agent Orchestrator | *FastAPI, LangGraph, FAISS, SQLite, WebSockets, SSE*

- Built a multi-agent **customer support investigation assistant** that combines customer data, API responses and knowledge-base documents to investigate issues and generate validated responses from multiple sources.
- **Live:** nexusgraph-multiagent-orchestrator.onrender.com
- **GitHub:** <https://github.com/Deeps72-ux/nexusgraph-multiagent-orchestrator>

OmniVoice – Streaming Voice Agent | *Python, FastAPI, LLMs, Streaming*

- Built a real-time voice agent with streaming responses, demonstrating low-latency conversational AI and asynchronous backend processing.
- **Live:** omnivoice-streaming-agent.onrender.com
- **GitHub:** <https://github.com/Deeps72-ux/omnivoice-streaming-agent>

DocuMind – Agentic RAG | *Python, RAG, LangGraph, Vector Search*

- Built a document QA application with document ingestion, vector search and agentic retrieval, demonstrating practical RAG implementation.
- **Live:** documind-agentic-rag-xuz4.onrender.com
- **GitHub:** github.com/Deeps72-ux/documind-agentic-rag

ProposalCraft – Agentic Proposal Generator | *FastAPI, LangGraph, Document Processing, LLMs*

- Built an application that converts RFP requirements into structured proposal documents, demonstrating LLM workflows and document generation.
- **Live:** proposalcraft-agent.onrender.com
- **GitHub:** github.com/Deeps72-ux/proposalcraft-agent

Certifications

Microsoft Certified: Azure Developer Associate (AZ-204) – Certification ID: **E83442-7985BF**; Expires: **30 Sep. 2027**
Microsoft Certified: Azure Fundamentals (AZ-900) – Certification ID: **3672EB-85BFR0**; **Does not expire**